

MEASURE THEORY

1. OUTER MEASURES

Definition 1.1. An *outer measure* on a set X is a function $\mu : 2^X \rightarrow [0, \infty]$ which satisfies the following conditions:

(1) Countably sub-additivity: if $A_1, A_2, \dots \subset X$ then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i).$$

(2) Monotonicity: If $A \subset B \subset X$ then $\mu(A) \leq \mu(B)$.

(3) Empty set is null: $\mu(\emptyset) = 0$.

We say that $A \subset X$ is μ -measurable if

$$\mu(B) = \mu(B \cap A) + \mu(B \cap A^c) \quad \text{for all } B \subset X.$$

Note that, by sub-additivity, this is equivalent to

$$\mu(B) \geq \mu(B \cap A) + \mu(B \cap A^c) \quad \text{for all } B \subset X.$$

Theorem 1.2. The set of all measurable subsets forms a σ -algebra on X . Moreover, μ is countably-additive (a measure) on X for this σ -algebra.

Proof. Observe that $\emptyset$ is measurable since for $B \subset X$ we have

$$\mu(B) = 0 + \mu(B \cap X) = \mu(\emptyset) + \mu(B \cap X) = \mu(B \cap \emptyset) + \mu(B \cap \emptyset^c).$$

Also A is measurable then A^c also is by symmetry.

We first show that the set of measurable sets is closed under finite intersection. Thus suppose that $A_1, A_2, \dots \subset X$ are measurable and $B \subset X$.

As A_1 is measurable, we obtain that

$$\mu(B \cap A_2^c) = \mu(B \cap A_2^c \cap A_1) + \mu(B \cap A_2^c \cap A_1^c)$$

and

$$\mu(B \cap A_2) = \mu(B \cap A_2 \cap A_1) + \mu(B \cap A_2 \cap A_1^c).$$

Now we add these two equations and use the inequality (which follows from subadditivity and the identity $A_1 \cup A_2 = (A_1^c \cap A_2) \cup (A_1 \cap A_2^c) \cup (A_1 \cap A_2)$)

$$\mu(B \cap A_2^c \cap A_1) + \mu(B \cap A_2 \cap A_1) + \mu(B \cap A_2 \cap A_1^c) \geq \mu(B \cap (A_1 \cup A_2))$$

to get that

$$\mu(B \cap A_2^c) + \mu(B \cap A_2) \geq \mu(B \cap (A_1 \cap A_2)) + \mu(B \cap A_1^c \cap A_2^c).$$

Measurability of A_2 means that the left hand side is $\mu(B)$, thus

$$\mu(B) \geq \mu(B \cap (A_1 \cap A_2)) + \mu(B \cap (A_1 \cup A_2)).$$

The reverse inequality holds from sub-additivity, thus $A_1 \cup A_2$ is measurable.

Thus we have shown that the measurable sets form an algebra. To show that they form a σ -algebra, it thus remains to show that a *disjoint* union of measurable sets is measurable. Thus suppose $A_1, A_2, \dots$ are disjoint measurable sets and $B \subset X$. We claim that

$$\mu \left(B \cap \left(\bigcup_{i=1}^n A_i \right) \right) = \mu(B \cap A_1) + \dots + \mu(B \cap A_n).$$

We prove this by induction as follows. Suppose it holds for n . Then by measurability of A_{n+1} we have that

$$\begin{aligned} \mu \left(B \cap \left(\bigcup_{i=1}^{n+1} A_i \right) \right) &= \mu \left(B \cap \left(\bigcup_{i=1}^{n+1} A_i \right) \cap A_{n+1} \right) + \mu \left(B \cap \left(\bigcup_{i=1}^{n+1} A_i \right) \cap A_{n+1}^c \right) \\ &= \mu(B \cap A_{n+1}) + \mu \left(B \cap \left(\bigcup_{i=1}^n A_i \right) \right) \end{aligned}$$

which completes the induction step (the last equation uses disjointness of the A_i). Now by monotonicity of μ we get

$$\mu \left(B \cap \left(\bigcup_{i=1}^{\infty} A_i \right) \right) \geq \lim_{n \rightarrow \infty} \mu \left(B \cap \left(\bigcup_{i=1}^n A_i \right) \right) = \sum_{i=1}^{\infty} \mu(B \cap A_i).$$

The reverse inequality holds from sub-additivity, thus we have shown that

$$\mu \left(B \cap \left(\bigcup_{i=1}^{\infty} A_i \right) \right) = \sum_{i=1}^{\infty} \mu(B \cap A_i).$$

In particular, for $B = X$ this shows that μ is countably-additive on the measurable sets. It remains to show that this countable union is measurable. Observe that since finite unions are measurable we have

$$\begin{aligned} \mu(B) &= \mu \left(B \cap \left(\bigcup_{i=1}^n A_i \right) \right) + \mu \left(B \cap \left(\bigcup_{i=1}^n A_i \right)^c \right) \\ &\geq \mu \left(B \cap \left(\bigcup_{i=1}^n A_i \right) \right) + \mu \left(B \cap \left(\bigcup_{i=1}^{\infty} A_i \right)^c \right). \end{aligned}$$

Now letting $n \rightarrow \infty$ and using countable sub-additivity, the proof is complete. $\square$

Example 1.3 (Lebesgue outer measure). If $I \subset \mathbb{R}$ is a non-empty bounded interval, we let $\text{len}(I) = \sup(I) - \inf(I)$ and $\text{len}(\emptyset) = 0$. Note I could be open, closed or half open. If $\mathcal{I}$ is a countable set of bounded intervals of $\mathbb{R}$, then we denote $\text{len}(\mathcal{I}) = \sum_{I \in \mathcal{I}} \text{len}(I)$. We say that $\mathcal{I}$ covers B if

$$B \subset \bigcup_{I \in \mathcal{I}} I.$$

Finally, denote the Lebesgue outer-measure of $B \subset \mathbb{R}$ to be

$$\lambda(B) = \inf_{\mathcal{I} \text{ covers } B} \text{len}(\mathcal{I}).$$

Let us now show that any bounded interval J is λ -measurable. To see this, let $B \subset \mathbb{R}$ be any subset and suppose that $\mathcal{I}$ covers B such that $\text{len}(\mathcal{I}) < \lambda(B) + \epsilon$. We can write $J^c = J_1 \cup J_2$ where J_1 and J_2 are disjoint (unbounded) intervals. Let $\mathcal{I}_1 = \{I \cap J \mid I \in \mathcal{I}\}$ and

$$\mathcal{I}_2 = \{I \cap J_m \mid I \in \mathcal{I}, m \in \{1, 2\}\}.$$

Then $\mathcal{I}_1$ covers $B \cap J$ and $\mathcal{I}_2$ covers $B \cap J^c$. Observe that $\text{len}(\mathcal{I}_1) + \text{len}(\mathcal{I}_2) = \text{len}(\mathcal{I})$ since

$$\text{len}(I) = \text{len}(I \cap J_1) + \text{len}(I \cap J) + \text{len}(I \cap J_2).$$

Thus

$$\lambda(B \cap J) + \lambda(B \cap J^c) \leq \text{len}(\mathcal{I}_1) + \text{len}(\mathcal{I}_2) = \text{len}(\mathcal{I}) \leq \lambda(B) + \epsilon.$$

Letting $\epsilon \rightarrow 0$ shows that J is λ -measurable.

2. MONOTONE CLASSES

Definition 2.1. Let X be a set. A monotone class on X is a collection of subsets $\mathcal{A} \subset 2^X$ which satisfies the following properties.

- (1) $\emptyset, X \in \mathcal{A}$.
- (2) If $A_1, A_2, \dots \in \mathcal{A}$ is such that $A_1 \subset A_2 \subset \dots$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$.
- (3) If $A_1, A_2, \dots \in \mathcal{A}$ is such that $A_1 \supset A_2 \supset \dots$ then $\bigcap_{i=1}^{\infty} A_i \in \mathcal{A}$.

Recall that an algebra on X is a collection of subsets of X that is closed under finite intersections, finite unions, complements and contains $\emptyset$ and X .

Theorem 2.2 (Monotone Class Lemma). Let X be a set and suppose that $\mathcal{A}$ is an algebra of subsets of X . Then the σ -algebra generated by $\mathcal{A}$ is the smallest monotone class that contains $\mathcal{A}$.

Proof. Let $\mathcal{M}$ denote the smallest monotone class that contains $\mathcal{A}$. We must show that $\mathcal{M}$ is a σ -algebra. It is enough to show that $\mathcal{M}$ is closed under finite unions and complements. For $A \in \mathcal{A}$, let

$$\mathcal{B}_A = \{B \in \mathcal{M} \mid B \cup A \in \mathcal{M}\}.$$

Note that $\mathcal{A} \subset \mathcal{B}_A$ by definition of algebra.

We claim that $\mathcal{B}_A$ is a monotone class. To see this, suppose that $B_1 \subset B_2 \subset \dots$ are all in $\mathcal{B}_A$. Thus $B_i \cup A \in \mathcal{M}$. But $B_i \cup A$ is a nested sequence of sets, and so $\bigcup_i B_i \cup A \in \mathcal{M}$ and so $\bigcup_i B_i \in \mathcal{B}_A$. Likewise suppose that $B_1 \supset B_2 \supset \dots$ are all in $\mathcal{B}_A$. Thus $B_i \cup A \in \mathcal{M}$. As $B_i \cup A$ monotonically decreases to $A \cup \bigcap_i B_i$ we have that $A \cup \bigcap_i B_i \in \mathcal{M}$ and so $\bigcap_i B_i \in \mathcal{B}_A$.

Thus $\mathcal{B}_A$ is a monotone class containing $\mathcal{M}$, but is a subset of $\mathcal{M}$ by definition. Thus $\mathcal{M} = \mathcal{B}_A$. Now let

$$\mathcal{M}_0 = \{M_0 \in \mathcal{M} \mid M_0 \cup M \in \mathcal{M} \text{ for all } M \in \mathcal{M}\}.$$

We see that $\mathcal{A} \subset \mathcal{M}_0$ since $\mathcal{M} = \mathcal{B}_A$. Note also that $\mathcal{M}_0$ is a monotone class. Thus $\mathcal{M}_0 = \mathcal{M}$. Thus we have shown that $\mathcal{M}$ is closed under finite unions, and hence countable unions.

Now let $\mathcal{M}' = \{M \in \mathcal{M} \mid M^c \in \mathcal{M}\}$. Note that $\mathcal{A} \subset \mathcal{M}'$ and $\mathcal{M}'$ is a monotone class. Thus $\mathcal{M}' = \mathcal{M}$ and so $\mathcal{M}$ is closed under complements. Thus $\mathcal{M}$ is indeed a σ -algebra. □

Example 2.3. Let $\mathcal{A}$ denote the set of all finite unions of half-open intervals $\bigcup_{i=1}^n [a_i, b_i)$ for some $n \geq 0$. Then $\mathcal{A}$ is an algebra, but not a σ -algebra. The monotone class lemma says that the smallest monotone class containing $\mathcal{A}$ is the Borel σ -algebra.

Now let $\mathcal{U}$ be the smallest set containing $\mathcal{A}$ that is closed under countable nested unions. We claim that each element in $\mathcal{U}$ is either empty or has positive Lebesgue measure. Let $\mathcal{U}'$ denote all $U \in \mathcal{U}$ with this property (U is either empty or positive measure). Clearly $\mathcal{A} \subset \mathcal{U}'$. Now let $U_1, U_2, \dots \in \mathcal{U}'$ and $U = \bigcup_i U_i$. If U has zero measure, then each U_i has zero measure and so each U_i empty and so U is empty. Thus $U \in \mathcal{U}'$. Thus $\mathcal{U} = \mathcal{U}'$. This means that $\mathcal{U}$ is not the Borel σ -algebra. This shows that in the definition of monotone class,

we really do need closure under countable intersections as well otherwise the monotone class lemma is not true.

3. PRODUCT MEASURES AND FUBINI-TONELLI THEOREM

Let X, Y be measure spaces and let

$$\mathcal{R}(X, Y) = \{U \times V \mid U \subset X \text{ measurable}, V \subset Y \text{ measurable}\}$$

Denote the set of all rectangles with in $X \times Y$ with measurable projections onto the axes. Let μ and ν be two positive measures on X and Y respectively. Then we define the area

$$\text{Area}(U \times V) = \mu(U)\nu(V).$$

If $\mathcal{C} \subset \mathcal{R}$ is a countable collections of rectangles define the

$$\text{Area}(\mathcal{C}) = \sum_{C \in \mathcal{C}} \text{Area}(C).$$

We say that $\mathcal{C}$ covers $A \subset X \times Y$ if

$$A \subset \bigcup_{C \in \mathcal{C}} C.$$

Define the product outer-measure

$$(\mu \times \nu)(A) = \inf_{\mathcal{C} \text{ covers } A} \text{Area}(\mathcal{C}).$$

Proposition 3.1. Each element of $\mathcal{R}$ is $\mu \times \nu$ -measurable.

Proof. Let $R \in \mathcal{R}$ and let $B \subset X \times Y$. Let $\mathcal{C}$ be a countable collection of rectangles that covers B such that

$$\text{Area}(\mathcal{C}) \leq (\mu \times \nu)(B) + \epsilon.$$

Observe that the complement R^c can be written as a disjoint union of finitely many rectangles. More precisely, if $R = U \times V$ then

$$R^c = U^c \times V^c \cup U^c \times V \cup U \times V^c.$$

Call these three rectangles R_1, R_2, R_3 . Observe that any rectangle $C = C_1 \times C_2 \in \mathcal{C}$ we have that

$$\begin{aligned} & \text{Area}(C \cap R) + \text{Area}(C \cap R_1) + \text{Area}(C \cap R_2) + \text{Area}(C \cap R_3) \\ &= \mu(C_1 \cap U)\nu(C_2 \cap V) + \mu(C_1 \cap U^c)\nu(C_2 \cap V^c) + \mu(C_1 \cap U^c)\nu(C_2 \cap V) + \mu(C_1 \cap U)\nu(C_2 \cap V^c) \\ &= (\mu(C_1 \cap U) + \mu(C_1 \cap U^c))(\nu(C_2 \cap V) + \nu(C_2 \cap V^c)) \\ &= \mu(C_1)\nu(C_2) \\ &= \text{Area}(C). \end{aligned}$$

Thus if we let

$$\mathcal{C}_2 = \{C \cap R_i \mid i = 1, 2, 3 \text{ and } C \in \mathcal{C}\}$$

and

$$\mathcal{C}_1 = \{C \cap R \mid C \in \mathcal{C}\}$$

then

$$\text{Area}(\mathcal{C}_1) + \text{Area}(\mathcal{C}_2) = \text{Area}(\mathcal{C}).$$

Observe that $\mathcal{C}_1$ covers $B \cap R$ and $\mathcal{C}_2$ covers $B \cap R^c$ and so

$$(\mu \times \nu)(R \cap B) + (\mu \times \nu)(R^c \cap B) \leq \text{Area}(\mathcal{C}_1) + \text{Area}(\mathcal{C}_2) = \text{Area}(\mathcal{C}) \leq (\mu \times \nu)(B) + \epsilon.$$

By letting $\epsilon \rightarrow 0$, the proof is complete. $\square$

We let $\mathcal{B}_{X \times Y}$ denote the smallest σ -algebra on $X \times Y$ containing $\mathcal{R}$. Note that this Proposition and Theorem 1.2 says that $\mu \times \nu$ is a measure (countably sub-additive) on this σ -algebra.

Another convenient definition of $\mu \times \nu$ is as follows. Let $\mathcal{C}$ is a countable collection of rectangles, then define the positive function

$$F_{\mathcal{C}} : X \times Y \rightarrow [0, \infty]$$

to be

$$F_{\mathcal{C}}(a) = \sum_{C \in \mathcal{C}} \mathbb{1}_C(a).$$

Then notice that $\mathcal{C}$ covers A if and only if $\mathbb{1}_A \leq F_{\mathcal{C}}$. Now consider the function $X \rightarrow [0, \infty]$ given by

$$x \mapsto \int_Y F_{\mathcal{C}}(x, y) d\nu(y).$$

Let us show that this function is well defined and measurable. If $F_{\mathcal{C}} = \sum_{i=1}^{\infty} \mathbb{1}_{U_i \times V_i}$ then

$$\int_Y F_{\mathcal{C}}(x, y) d\nu(y) = \int_Y \mathbb{1}_{U_i}(x) \mathbb{1}_{V_i}(y) d\nu(y) = \sum_{i=1}^{\infty} \mathbb{1}_{U_i}(x) \nu(V_i)$$

by the monotone convergence theorem. Thus this function is indeed measurable (by being the countable sum of non-negative measurable functions). By applying the monotone convergence theorem again we have the useful Fubini-type equation

$$\int_X \int_Y F_{\mathcal{C}}(x, y) d\nu(y) d\mu(x) = \sum_{i=1}^{\infty} \mu(U_i) \nu(V_i) = \text{Area}(\mathcal{C}).$$

Proposition 3.2. If $R \in \mathcal{R}$ is a rectangle, then $(\mu \times \nu)(R) = \text{Area}(R)$.

Proof. Suppose that $\mathcal{C}$ covers R . Then we must have that

$$F_{\{R\}} = \mathbb{1}_R \leq F_{\mathcal{C}}.$$

Thus we must have that

$$\text{Area}(R) = \int_X \int_Y F_{\{R\}}(x, y) d\nu(y) d\mu(x) \leq \int_X \int_Y F_{\mathcal{C}}(x, y) d\nu(y) d\mu(x) = \text{Area}(\mathcal{C}).$$

As $\mathcal{C}$ was arbitrary, this shows that

$$(\mu \times \nu)(R) \geq \text{Area}(R).$$

The reverse inequality holds because $\{R\}$ covers R . $\square$

Theorem 3.3 (Fubini-Tonelli theorem for indicator functions). Suppose that μ and ν are σ -finite (on X and Y respectively). Let $B \in \mathcal{B}_{X \times Y}$. Then the mapping

$$x \mapsto \int_Y \mathbb{1}_B(x, y) d\nu(y)$$

is measurable and its integral over X is $(\mu \times \nu)(B)$, that is,

$$\int_X \int_Y \mathbb{1}_B(x, y) d\nu(y) d\mu(x) = (\mu \times \nu)(B) = \int_Y \int_X \mathbb{1}_B(x, y) d\mu(x) d\nu(y).$$

Proof. By a partitioning argument and the monotone convergence theorem, we can reduce to the case where μ and ν are both finite.

Let $\mathcal{M}$ denote the set of all $B \in \mathcal{B}_{X \times Y}$ for which this theorem is true. By the proposition above, $\mathcal{R} \subset \mathcal{M}$. Note that a finite union of rectangles may be written as a finite union of disjoint rectangles, thus the smallest algebra that contains $\mathcal{R}$ is a subset of $\mathcal{M}$. Thus, by the montone class lemma it will be enough to show that $\mathcal{M}$ is a montone class. That $\mathcal{M}$ is closed under countable nested unions follows from the monotone convergence theorem. To show that $\mathcal{M}$ is closed under countable nested intersections, we can use the fact that $\mathcal{M}$ is closed under complements, which uses the finiteness of μ and ν as follows. Indeed if $B \in \mathcal{M}$, then $\int_X \int_Y \mathbb{1}_{B^c}(x, y) d\nu(y) d\mu(x) = \int_X \int_Y (1 - \mathbb{1}_B(x, y)) d\nu(y) d\mu(x) = (\mu \times \nu)(X \times Y) - (\mu \times \nu)(B) = (\mu \times \nu)(B^c)$ and so $B^c \in \mathcal{M}$. $\square$

Example 3.4. The σ -finiteness assumption is necessary. For instance, let $X, Y = \mathbb{R}$ and let μ be the counting measure (not σ -finite) and let ν be the Lebesgue measure and let $B = \{(x, x) \mid x \in \mathbb{R}\}$ be the diagonal. Then

$$\int_X \int_Y \mathbb{1}_B(x, y) d\nu(y) d\mu(x) = \int_X 0 d\mu(x) = 0$$

and yet

$$\int_Y \int_X \mathbb{1}_B(x, y) d\mu(x) d\nu(y) = \int_Y 1 d\nu(y) = \infty.$$

By using the monotone convergence theorem, we get the Fubini-Tonelli theorem for positive measurable functions.

Theorem 3.5 (Tonelli theorem). Suppose that μ and ν are σ -finite (on X and Y respectively) and suppose that $f : X \times Y \rightarrow [0, \infty]$ is $\mathcal{B}_{X \times Y}$ -measurable. Then

$$x \mapsto \int_Y f(x, y) d\nu(y)$$

is measurable and

$$\int_X \int_Y f(x, y) d\nu(y) d\mu(x) = \int_{X \times Y} f(x, y) d(\mu \times \nu)(x, y) = \int_Y \int_X f(x, y) d\mu(x) d\nu(y).$$

Proof. Write $f = \sum_{i=1}^{\infty} c_i \mathbb{1}_{B_i}$ for some $B_i \in \mathcal{B}_{X \times Y}$ and $c_i \in [0, \infty)$. Now apply the previous theorem and the monotone convergence theorem. $\square$

Theorem 3.6. Suppose that μ and ν are σ -finite (on X and Y respectively) and suppose that $f : X \times Y \rightarrow [-\infty, \infty]$ is $\mathcal{B}_{X \times Y}$ -measurable. Suppose that¹

$$\int_{X \times Y} |f(x, y)| d(\mu \times \nu)(x, y) < \infty.$$

Then there exists a measurable set $X' \subset X$ that is μ -conull and such that

$$x \mapsto \int_Y |f(x, y)| d\nu(y) < \infty \quad \text{for all } x \in X'$$

and the map $X' \rightarrow X$ given by

$$x \mapsto \int_Y f(x, y) d\nu(y)$$

¹By Tonelli's theorem, we can replace this condition with an iterated integral for $|f|$.

is a well defined measurable map. Furthermore²

$$\int_X \int_Y f(x, y) d\nu(y) d\mu(x) = \int_{X \times Y} f(x, y) d(\mu \times \nu)(x, y) = \int_Y \int_X f(x, y) d\mu(x) d\nu(y).$$

Proof. As f is L^1 with respect to $\mu \times \nu$, we can write $f = f^+ - f^-$ where $f^+, f^- : X \times Y \rightarrow [0, \infty]$ and f^+, f^- are both L^1 . Thus by an application of Tonelli's theorem to both of these functions we have that

$$\int_X \int_Y f^\pm(x, y) d\nu(y) d\mu(x) < \infty$$

and so there must be a co-null set $X' \subset X$ so that

$$\int_Y f^\pm(x, y) d\nu(y) < \infty \quad \text{for all } x \in X'$$

and this quantity is a measurable function on X' . Thus

$$\begin{aligned} \int_{X'} \int_Y f(x, y) d\nu(y) d\mu(x) &= \int_{X'} \int_Y (f^+(x, y) - f^-(x, y)) d\nu(y) d\mu(x) \\ &= \int_{X'} \left(\int_Y f^+(x, y) d\nu(y) - \int_Y f^-(x, y) d\nu(y) \right) d\mu(x) \\ &= \int_{X'} \int_Y f^+(x, y) d\nu(y) d\mu(x) - \int_{X'} \int_Y f^-(x, y) d\nu(y) d\mu(x) \\ &= \int_X \int_Y f^+(x, y) d\nu(y) d\mu(x) - \int_X \int_Y f^-(x, y) d\nu(y) d\mu(x) \\ &= \int_{X \times Y} f^+(x, y) d(\mu \times \nu)(x, y) - \int_{X \times Y} f^-(x, y) d(\mu \times \nu)(x, y) \\ &= \int_{X \times Y} f(x, y) d(\mu \times \nu)(x, y) \end{aligned}$$

□

²The integrands are only almost everywhere well defined. For example the first integral is over X' rather than X , but as X' is co-null we can still sensibly integrate over X by extending the domain to whichever function on X we like.